

```
#include <stdio.h>    // printf(), fgets()
#include <stdlib.h>    // exit()
#include <string.h>
#include <unistd.h>    // fork(), execvp(), getpid(), getppid()
#include <sys/types.h> // pid_t
#include <sys/wait.h> // wait()

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if(pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }

    else if(pid == 0)
    {
        printf("\n---- Child Process ----\n");

        printf("Child PID : %d\n", getpid());

        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n\n", command);

        char *args[] = {command, NULL};

        execvp(command, args);

        printf("Command execution failed!\n");

        exit(1);
    }

    else
    {
        printf("\n---- Parent Process ----\n");

        printf("Parent PID : %d\n", getpid());

        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

```

Last login: Mon Aug  3 21:12:25 on ttys000
(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop
vishalkarnati@VISHALS-MacBook-Air Desktop % nano command_executor.c
vishalkarnati@VISHALS-MacBook-Air Desktop % gcc command_executor.c -o command_executor
vishalkarnati@VISHALS-MacBook-Air Desktop % ./command_executor
Enter a Linux command: ls

----- Parent Process -----
Parent PID : 5236
Child PID : 5237

----- Child Process -----
Child PID : 5237
Parent PID : 5236
Executing command: ls

ai.png                               my-app
airline-seat-selection               myreact
certificates                         mysql-9.7.1-macos15-arm64.dmg
CFAI                                 node project.png
c1g                                  os
command_executor                     OSSP
command_executor.c                   placement_predict_50k Dataset.csv
datastru                             Practical_Week2.docx
DSA                                  productboard
dsa practical                         Projects_DATABASE_SYSTEMS_ENGINEERING_AND_DISTRIBUTED_BACKENDDEV.pdf
dsa2                                  python CFAI
EDA_Report (1) (2).pdf               reactpract
experiment2 (3).ipynb                Recursion
fedf pract                           routing
harireact                            routing2
harireactcopy                        run node.png
harry                                Screenshot 2024-07-28 at 2.45.58 PM.png
html project                          Screenshot 2024-08-03 at 9.13.10 PM.png
IMG_20240628_135101_644.webp         script
java                                  seat selection
lari game                             sort
ml                                     stach
mock-test-project                    studentboard
my_skills                             week2

Child process completed.
vishalkarnati@VISHALS-MacBook-Air Desktop % ./command_executor
Enter a Linux command: pwd

----- Parent Process -----
Parent PID : 5239
Child PID : 5240

----- Child Process -----
Child PID : 5240
Parent PID : 5239
Executing command: pwd

/Users/vishalkarnati/Desktop

Child process completed.

```

```
vishalkarnati@VISHALs-MacBook-Air Desktop % ./command_executor
Enter a Linux command: date

----- Parent Process -----
Parent PID : 5241
Child PID : 5242

----- Child Process -----
Child PID : 5242
Parent PID : 5241
Executing command: date

Mon Aug 3 21:14:23 IST 2026

Child process completed.
vishalkarnati@VISHALs-MacBook-Air Desktop % ./command_executor
Enter a Linux command: whoami

----- Parent Process -----
Parent PID : 5243
Child PID : 5245

----- Child Process -----
Child PID : 5245
Parent PID : 5243
Executing command: whoami

vishalkarnati

Child process completed.
vishalkarnati@VISHALs-MacBook-Air Desktop % ./command_executor
Enter a Linux command: hostname

----- Parent Process -----
Parent PID : 5246
Child PID : 5247

----- Child Process -----
Child PID : 5247
Parent PID : 5246
Executing command: hostname

VISHALs-MacBook-Air.local

Child process completed.
vishalkarnati@VISHALs-MacBook-Air Desktop % ./command_executor
Enter a Linux command: uname

----- Parent Process -----
Parent PID : 5248
Child PID : 5249

----- Child Process -----
Child PID : 5249
Parent PID : 5248
Executing command: uname

Darwin

Child process completed.
vishalkarnati@VISHALs-MacBook-Air Desktop % █
```